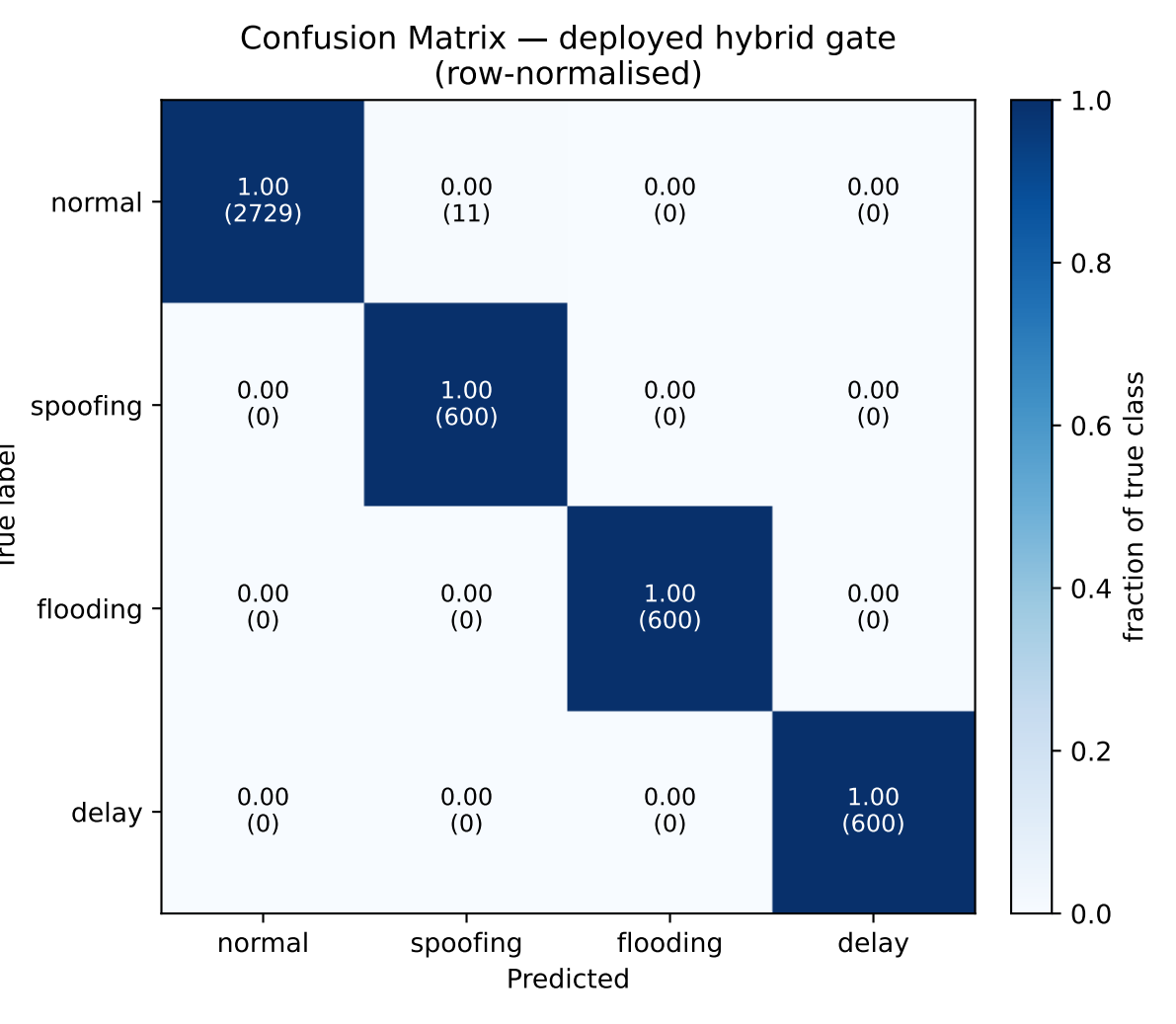
```
tb3_security — Offline Detection Evaluation
=====================================================

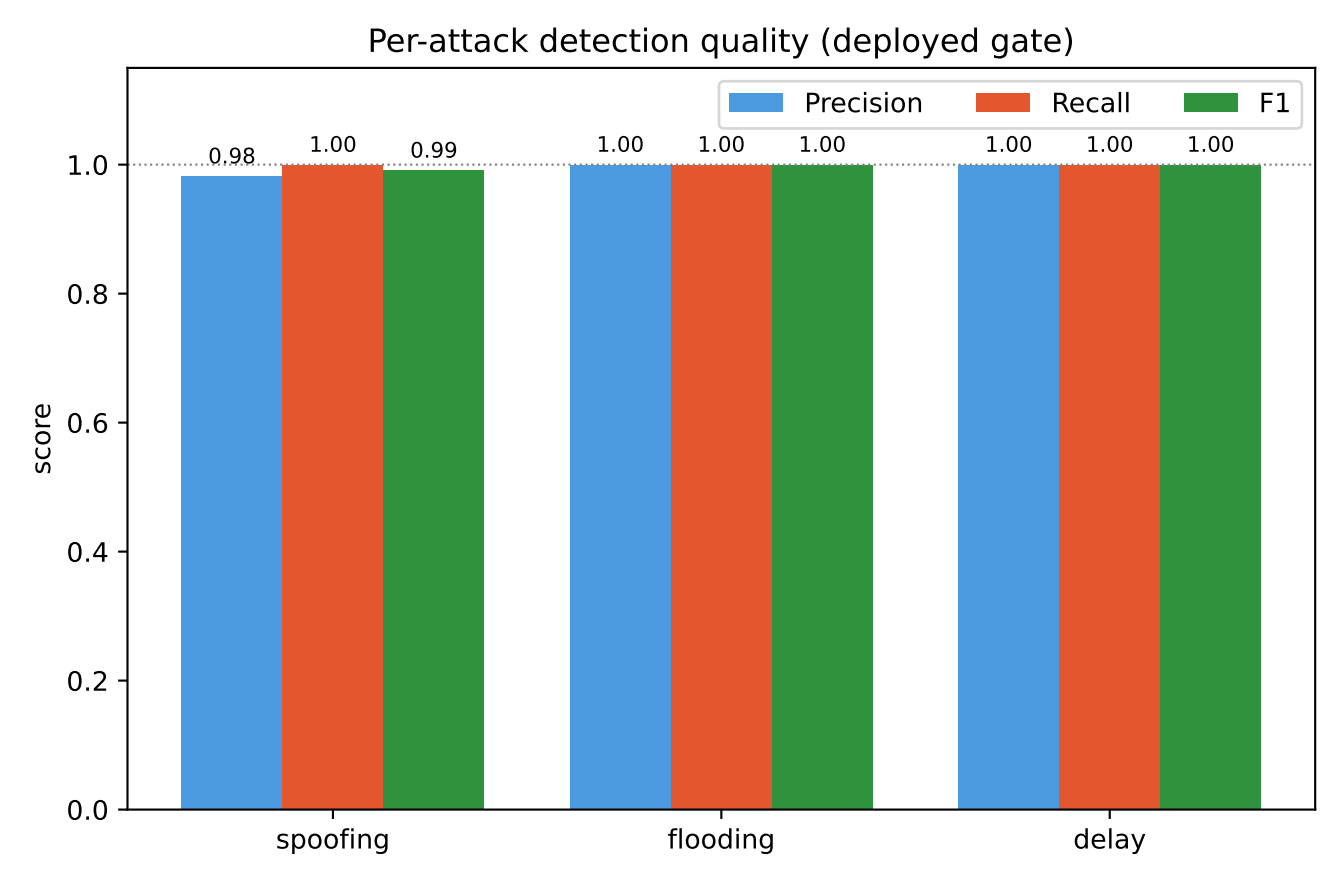
Classification samples : 4540  (normal=2740, attacks=1800)
Overall accuracy       : 0.998
Normal false-pos rate  : 0.004  (deployed hybrid gate; previous AND gate 0.004)

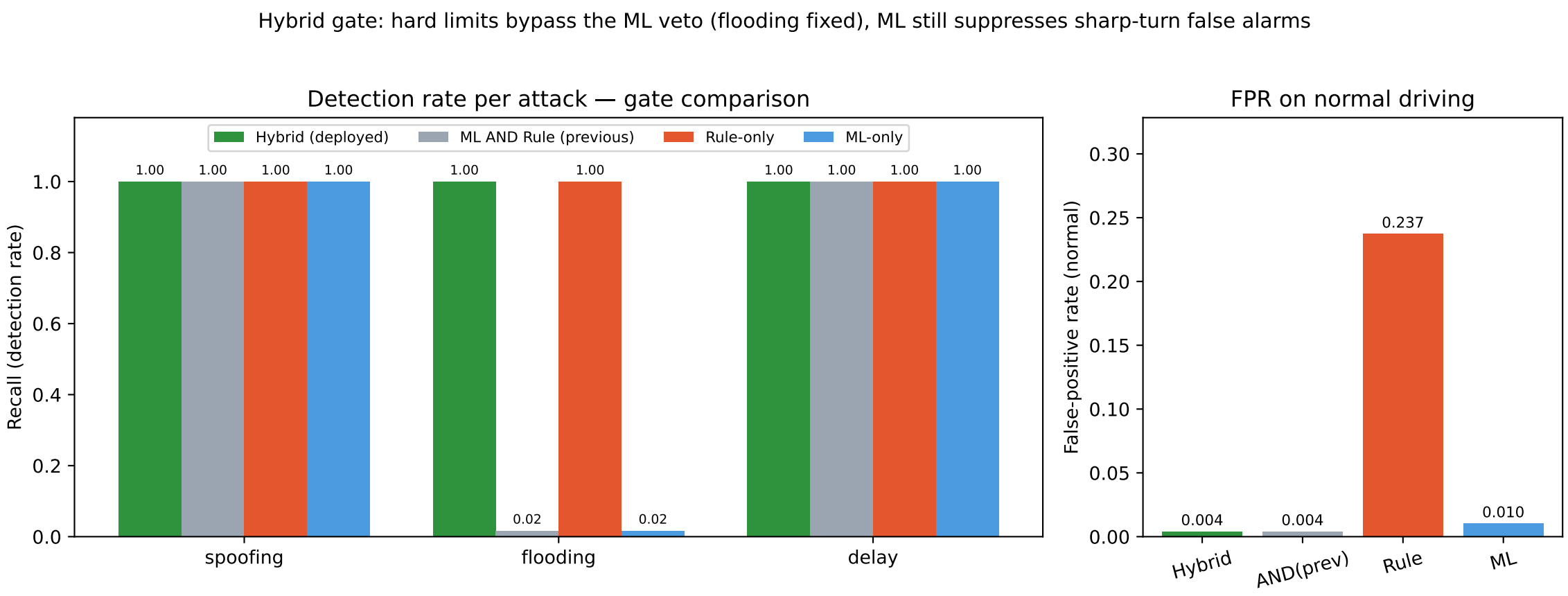
Per-attack recall (detection rate):
   spoofing   deployed=1.00   previous-AND=1.00   rule-only=1.00
   flooding   deployed=1.00   previous-AND=0.02   rule-only=1.00
   delay      deployed=1.00   previous-AND=1.00   rule-only=1.00

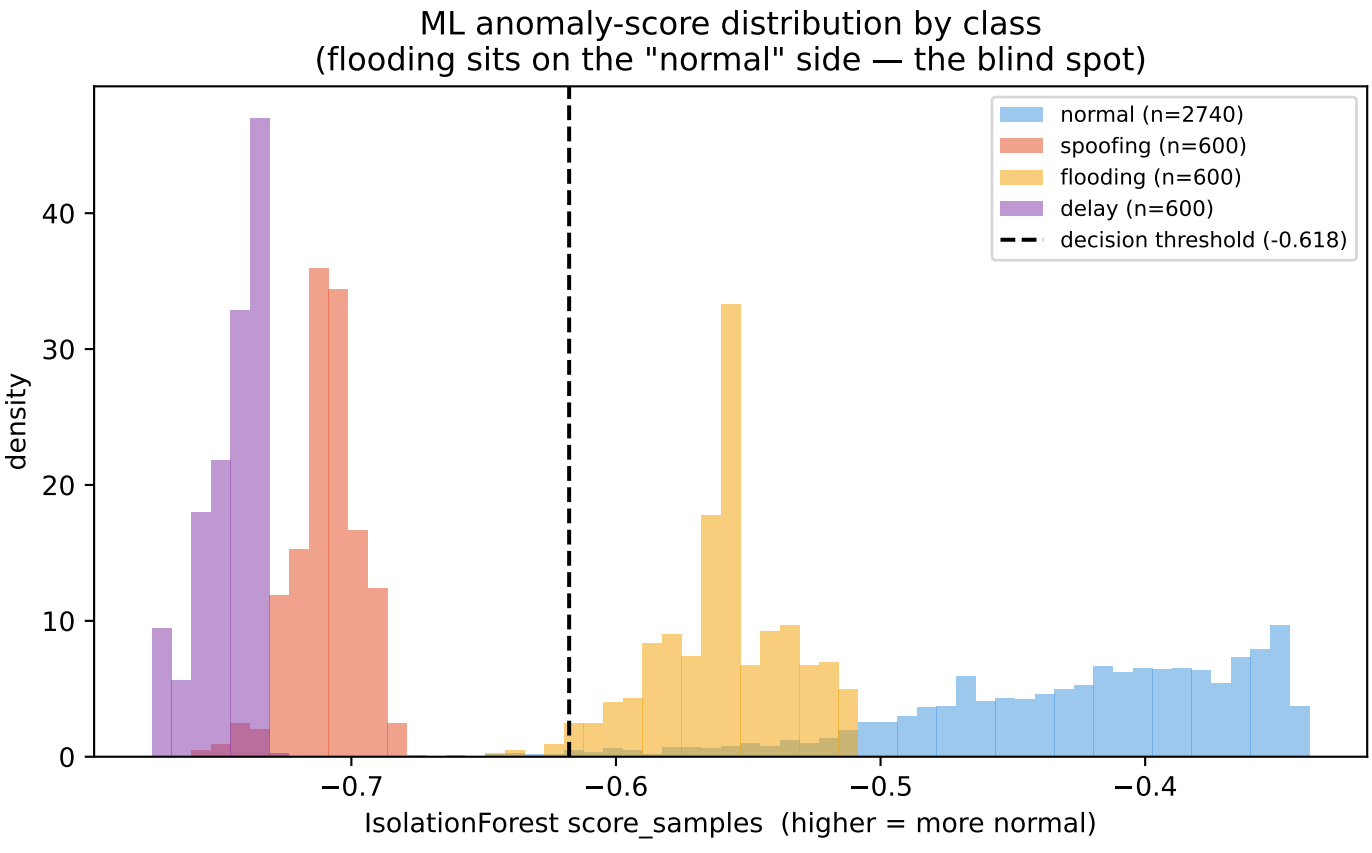
Detection latency (temporal scenario):
   episode #1  spoofing  -> 0.00s
   episode #2  flooding  -> 0.00s
   episode #3  delay     -> 0.00s
   episode #4  spoofing  -> 0.00s
   episode #5  flooding  -> 0.00s
   episode #6  delay     -> 0.00s
   episode #7  spoofing  -> 0.00s
   episode #8  flooding  -> 0.00s
   episode #9  delay     -> 0.00s
   episode #10 spoofing  -> 0.00s

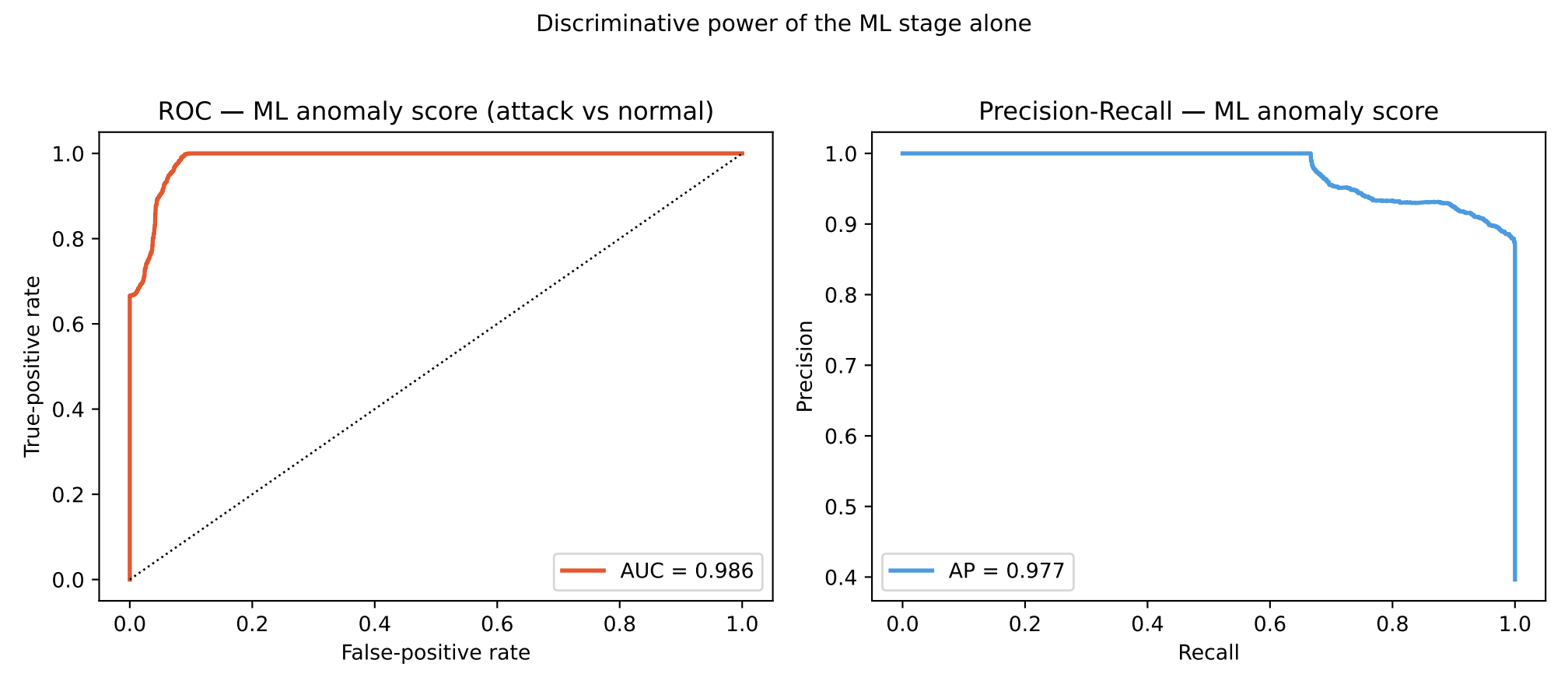
Key finding:
  * Previous gate  ML AND (sliding OR threshold)  missed flooding:
    the IsolationForest score is flat in publish_rate (trained on
    rate 1-14), so ML=normal vetoed the correct rule alarm.
  * Fix (deployed):  hard_limit OR (ML AND (sliding OR velocity)).
    Flood rate / delay are unambiguous and bypass the ML veto;
    ML still gates the velocity check, suppressing false alarms on
    legitimate sharp turns (angular_z up to 2.84).
```

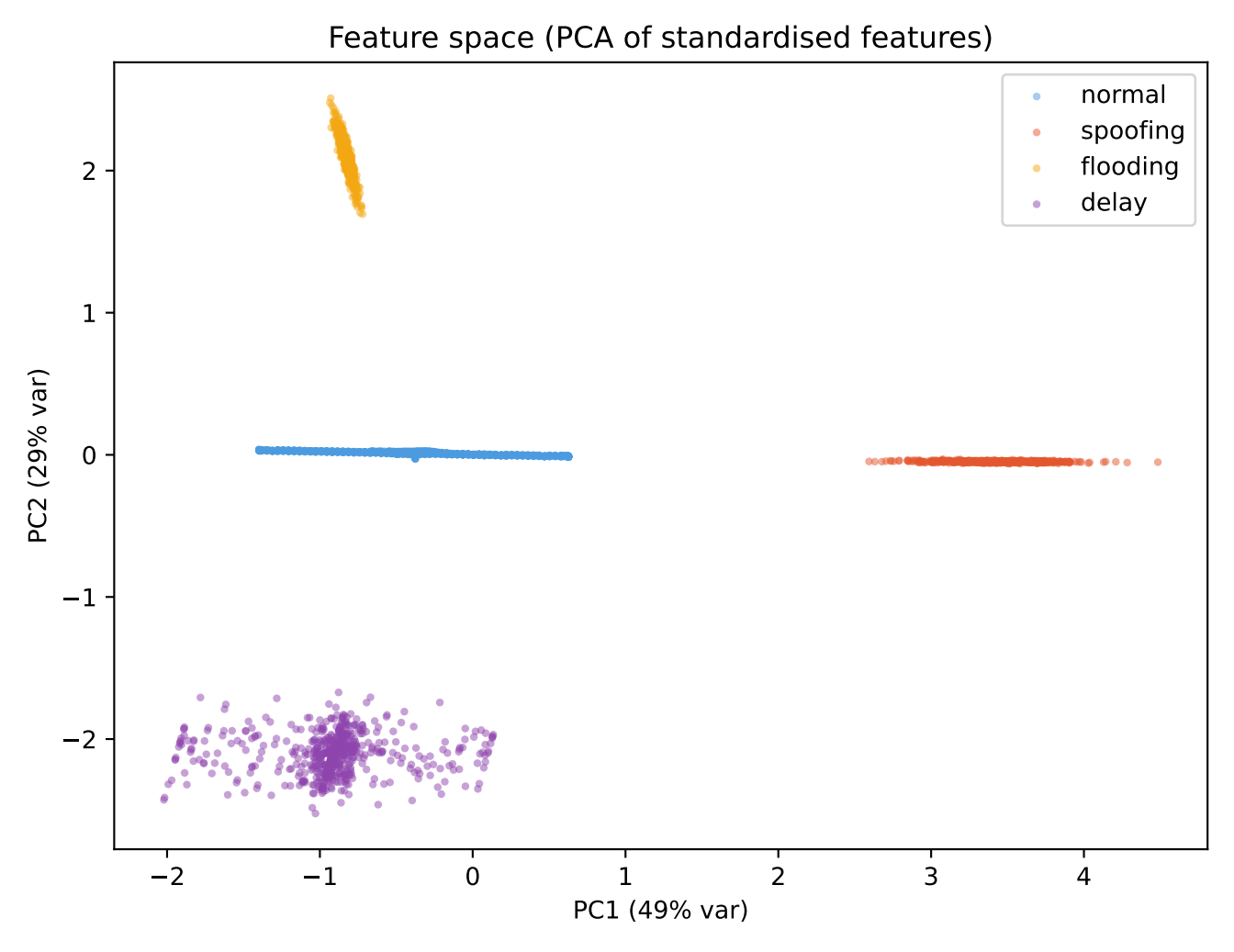

Confusion Matrix — deployed hybrid gate
(row-normalised)
normal
spoofing
flooding
delay
True label
1.00
(2729)
0.00
(11)
0.00
(0)
0.00
(0)
0.00
(0)
1.00
(600)
0.00
(0)
0.00
(0)
0.00
(0)
0.00
(0)
1.00
(600)
0.00
(0)
0.00
(0)
0.00
(0)
0.00
(0)
1.00
(600)
normal
spoofing
flooding
delay
Predicted
fraction of true class
1.0
0.8
0.6
0.4
0.2
0.0

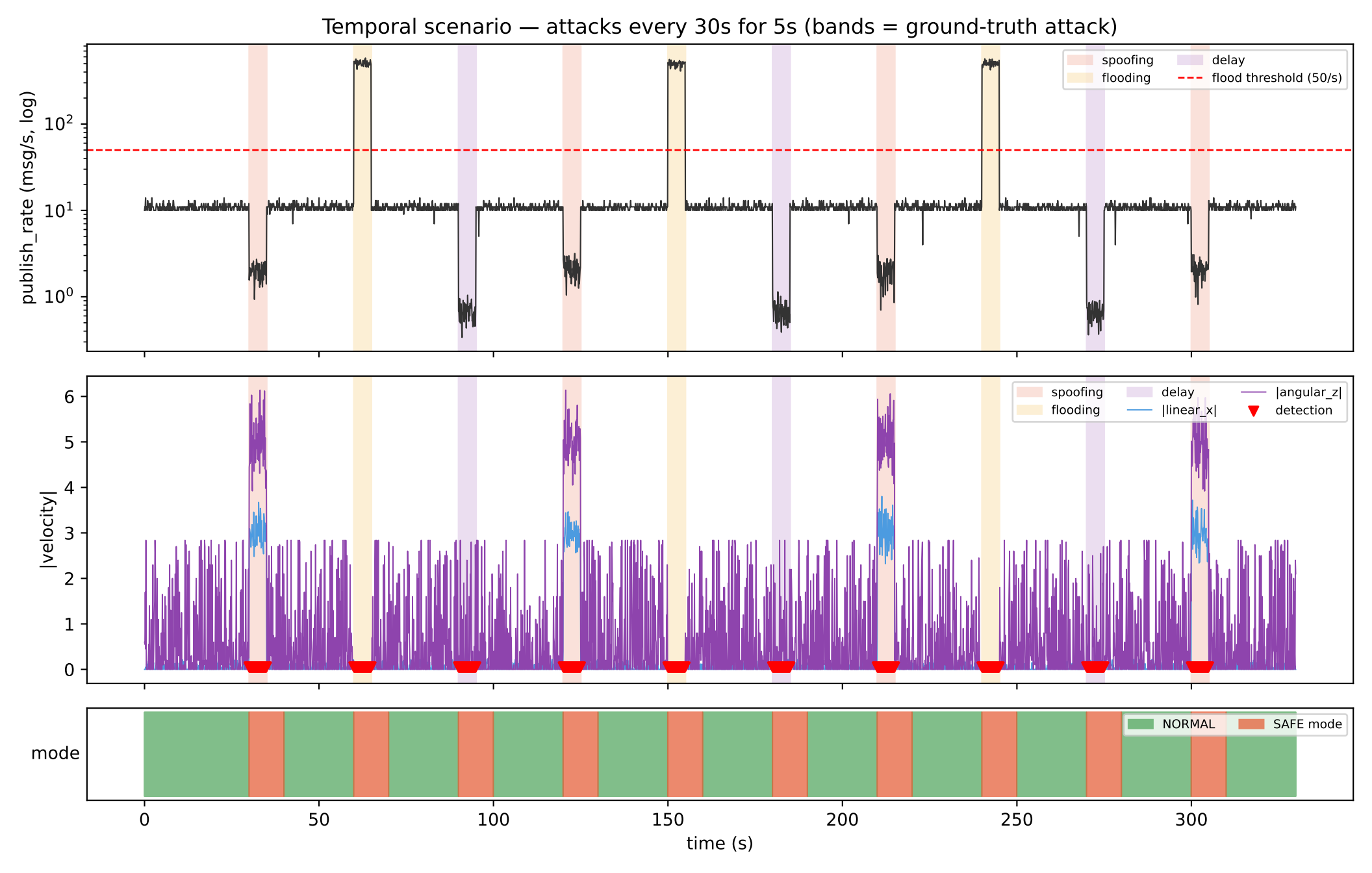

Per-attack detection quality (deployed gate)
Precision
Recall
F1
score
0.0
0.2
0.4
0.6
0.8
1.0
0.98
1.00
0.99
1.00
1.00
1.00
1.00
1.00
1.00
spoofing
flooding
delay

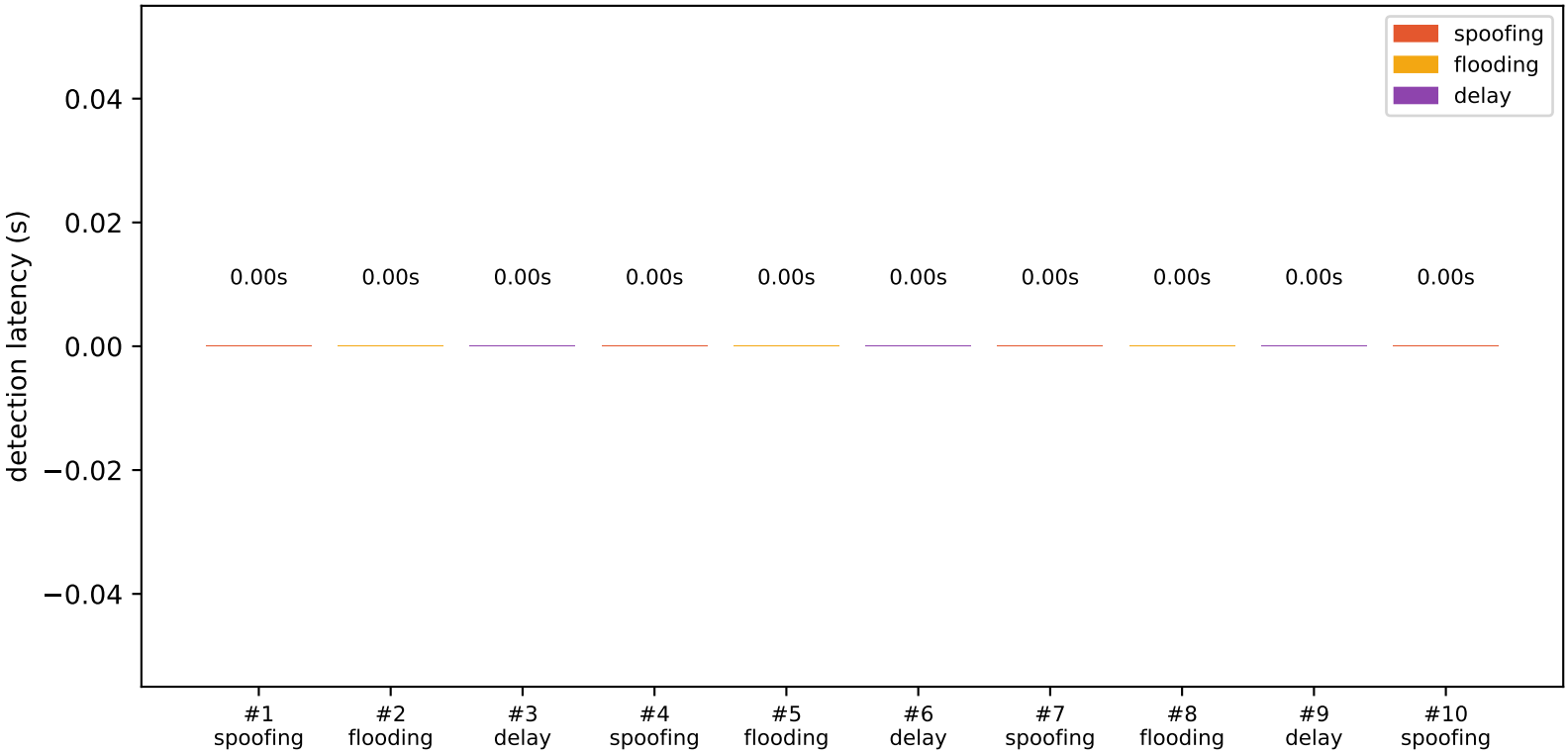

Hybrid gate: hard limits bypass the ML veto (flooding fixed), ML still suppresses sharp-turn false alarms
Detection rate per attack — gate comparison
Hybrid (deployed)
ML AND Rule (previous)
Rule-only
ML-only
Recall (detection rate)
1.00
0.02
spoofing
flooding
delay
FPR on normal driving
False-positive rate (normal)
0.004
0.237
0.010
Hybrid
AND(prev)
Rule
ML


ML anomaly-score distribution by class
(flooding sits on the "normal" side — the blind spot)
normal (n=2740)
spoofing (n=600)
flooding (n=600)
delay (n=600)
decision threshold (-0.618)
density
0
10
20
30
40
−0.7
−0.6
−0.5
−0.4
IsolationForest score_samples (higher = more normal)


Discriminative power of the ML stage alone
ROC — ML anomaly score (attack vs normal)
True-positive rate
False-positive rate
AUC = 0.986
Precision-Recall — ML anomaly score
Precision
Recall
AP = 0.977


Feature space (PCA of standardised features)
normal
spoofing
flooding
delay
PC1 (49% var)
PC2 (29% var)
−2
−1
0
1
2
3
4


Temporal scenario — attacks every 30s for 5s (bands = ground-truth attack)
publish_rate (msg/s, log)
spoofing
flooding
delay
flood threshold (50/s)
|velocity|
|angular_z|
|linear_x|
detection
mode
NORMAL
SAFE mode
time (s)


Time-to-detection per attack episode (0-height + MISSED = never detected)
spoofing
flooding
delay
detection latency (s)
0.04
0.02
0.00
−0.02
−0.04
0.00s
0.00s
0.00s
0.00s
0.00s
0.00s
0.00s
0.00s
0.00s
0.00s
#1 spoofing
#2 flooding
#3 delay
#4 spoofing
#5 flooding
#6 delay
#7 spoofing
#8 flooding
#9 delay
#10 spoofing